



Santa Cruz
Compounding
Academy

Basic Non-Sterile Compounding

for Pharmacists and Pharmacy Technicians: Day 3

Speaker: Lcdo. Jorge L. Reyes Quiñones, RPh

Institution: Santa Cruz Compounding Academy

Contact Hours: 6.0 | **Application Level:** 1 (Introductory)

Colegio de Farmacéuticos de Puerto Rico · Continuing Education Division
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ACPE Provider Number: 0151

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Provider Number	0151
Contact Hours (Day 3)	6.0 · Full course: 18 hours / 1.8 CEUs
Knowledge Level	1 (Introductory) — Application-Based
Deadline to claim credits	45 days from the event
Validation	cfpr.org + CPE Monitor

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Objectives — Day 3

Upon completion of the CPE, the participant will be able to:

- 1 Discuss the formulation and describe the use of equipment and devices for pediatric formulations.
- 2 Prepare two pediatric formulations.
- 3 Review the calculations and procedures used to prepare selected veterinary formulations.
- 4 Apply pharmaceutical calculations to accurately determine the ingredient quantities required for compounding Bioidentical Hormone Replacement Therapy (BHRT) preparations.
- 5 Use the appropriate devices and equipment in preparing Bioidentical Hormone Replacement Therapy (BHRT) preparations, in accordance with established safety and quality standards.
- 6 Prepare Bioidentical Hormone Replacement Therapy (BHRT) preparations following the appropriate formulation and compounding procedures to ensure product quality and consistency.

Source: ACPE Standards 2015 · Revised Bloom's Taxonomy · USP <795> 2023 · FDA-CVM AMDUCA

Instructional Methodology

Time distribution — Day 3 (6 hours)

- **Audiovisual lecture:** pediatrics, veterinary, BHRT.
- **Active learning:** 4 pop questions + 4 clinical cases (patient / animal).
- **Hands-on laboratory:** 4 practice formulations (BHRT cream, Enalapril suspension, veterinary Gabapentin suspension, extemporaneous Synthroid 25 mcg/mL).

Block	Duration
Review of Days 1 and 2 + Rules	60 min
Module 1: Pediatrics (Obj 1 + 2)	60 min
Module 2: Veterinary (Obj 3)	60 min
Module 3: BHRT (Obj 4)	60 min
Laboratory: 4 practice formulations	90 min
Post-test + Activity Evaluation + closing	30 min

Review — Days 1 and 2

Visual recap of the previous modules

DAY 1

- 503A vs. 503B (DQSA 2013).
- USP <795> Nov 2023: BUDs (14/35/90/180).
- USP <800>: HD handling + PPE + engineering.
- PR Act 247-2004 + Regulation 156.
- Calculations: potency, DV, fill weight.
- Lab: 100 SR capsules ($\pm 7.5\%$).

DAY 2

- Suppositories: DF + cocoa butter, PEG, Witepsol.
- Rectal / vaginal / urethral anatomy.
- Fusion and molding (weight control $\pm 5\%$).
- Dermatologicals: vehicles + devices.
- Pediatrics: intro to vehicles and 35-day BUD.
- Lab: 12 suppositories + 2 derm formulations.

Source: USP <795> 2023 · DQSA 2013 · PR Act 247-2004

Needs Assessment

Needs analysis — why this course?

Why does it matter?	What to expect after the course?	What the participant should know	Application Activity
~8.7M veterinary compounding Rx/year (FDA-CVM 2023)	Veterinary preparation under AMDUCA + a valid VCPR	Species, mg/kg, chews, feline TD, Schedule IV log	Veterinary suspension + veterinary-use label
2.5M BHRT Rx/year in the U.S. (PCCA 2023)	Patient-specific BHRT formulation	Bi-est ratios, troches, Topi-CLICK, bioidentical hormones	BHRT batch calculation in Topi-CLICK / Microclick
70–80% of medications have no FDA pediatric formulation	Reproducible pediatric suspensions + counseling	Ora-Plus/Sweet vehicles, BUDs, pediatric oral syringe	2 pediatric suspensions + caregiver counseling
FDA shortage + excipient allergy — 503A exceptions	Regulatorily correct compounding in BHRT/veterinary	Shortage exceptions, allergenic excipient, USP does not guarantee efficacy	Documented clinical case with regulatory justification

Source: NABP 2023 · ACPE Needs Assessment Guidelines · USP <795>/<800> · PR Act 247-2004

Activity Planning

Objectives · Practices · Final product as evidence

Objective	Practice / Experience 1	Practice / Experience 2	Final Product (Application Activity)
1. Discuss pediatric formulations + equipment	Ora-Plus/Sweet/Blend selection	Oral syringe calibration + BUDs	Documentation of the selected vehicle
2. Prepare practice oral formulations (LAB)	Enalapril 1 mg/mL — 60 mL	Veterinary Gabapentin 100 mg/mL — 1.5 mL q8h	2 suspensions + label + syringe
3. Review veterinary calculations	mg/kg dose by species	30-day canine phenobarbital batch	Veterinary calculation report
4. Apply BHRT pharmaceutical calculations	Bi-est E3/E2 ratio	CoA adjustment + batch calculation	BHRT calculation report
5. Use BHRT devices and equipment	Topi-CLICK / Microclick calibration	Container / troche mold selection	Documented click verification
6. Prepare hormonal preparations (BHRT/thyroid)	Bi-est 80/20 2.5 mg/g / Testosterone 3 mg/g cream	Synthroid 25 mcg/mL extemporaneous (tablet-to-liquid conversion)	Hormonal products + Act 247 label

Source: ACPE Standards 2015 · Bloom's Taxonomy (Application Level) · USP <795>/<800>

Competencies Assessed

Apply equally to pharmacists, technicians and students

PHARMACISTS · TECHNICIANS · STUDENTS — Same 6 competencies

1. Foundational knowledge
2. Patient-centered care
3. Medication use system
4. Problem solving
5. Innovation
6. Professionalism

Post-test $\geq 70\%$ required — uniform assessment for all profiles

Pediatric Formulations (1/2)

Discussion and pediatric vehicles (Obj 1)

Persistent challenge: 70–80% of medications have no FDA-approved pediatric formulation (AAP 2023).

Validated vehicles

Ora-Plus®: aqueous suspending agent, pH 4.2, preserved (potassium sorbate).

Ora-Sweet® / Ora-Sweet SF®: sweetened vehicle. SF is sugar-free, for diabetics.

Ora-Blend®: Ora-Plus + Ora-Sweet 1:1, premixed.

Xanthan gum 0.1–0.5%: stable shear-thinning gel.

CMC 0.5–1%: controlled viscosity, wide pH range.

Flavorings + AAP rules

Sucralose / Stevia: preferred over saccharin in children <2 years.

FlavoRx® / PCCA: strawberry, vanilla, orange, grape, cherry, bubble gum.

⚠ **Sorbitol:** ≤0.5 g/dose (osmotic diarrhea).

⚠ **Ethanol:** <0.5% in children <6 years.

⚠ **Aspartame:** avoid in phenylketonuria (PKU).

Source: USP <795> 2023 · AAP 2023 · PCCA Pediatric · Allen's · Ora-Plus PI

Pediatric Formulations (2/2)

Pediatric equipment and devices

Equipment / Device	Main use
Oral syringes (0.5–20 mL)	Graduated in mL (NEVER tsp/tbsp) — ISMP Safe Practice
Calibrated dosing droppers	Volumes <1 mL in infants; dispense into the cheek
Amber bottle + dosing adaptor	Protects from light; allows syringe use without spilling
Porcelain / glass mortar	Trituration (porcelain) / levigation (glass)
Class I/II analytical balance	± 0.1 mg, daily NIST calibration, log
Volumetric pipettes / graduated cylinders	Precise oral solutions (q.s. to volume)

Pediatric BUDs, USP <795> 2023: Aqueous oral without preservative = 14 days (ref.) · Preserved aqueous oral = 35 days · Nonaqueous solution = 90 days. Always use the shorter BUD between USP and published stability data.

Source: USP <795> 2023 · ISMP Pediatric Safety · AAP · PCCA · ASHP

Pediatric Compounding — Process

Describe the formulation + caregiver instructions (Obj 1)

Process steps

1. Verify the Rx and the patient's weight.
2. Calculate the dose: mg/kg/day.
3. Select the vehicle: Ora-Sweet®, Ora-Plus®, syrup, flavored.
4. Weigh the CoA-adjusted API.
5. Levigation + incorporation into the vehicle.
6. Shake and verify uniformity.
7. Label, BUD, oral syringe / dropper.

Caregiver instructions

- Shake well before each dose (suspensions).
- Use a calibrated oral syringe / dropper.
- Exact dose — never household spoons.
- Storage: refrigerate 2–8°C per the BUD.
- Discard date (BUD) on the label.
- Wash the syringe after each use.
- Report adverse reactions to the pharmacist.

Source: USP <795> 2023 · PCCA Pediatric · AAP 2023 · Allen's

QUICK QUESTION #1 · Learning Check

Pediatric — BUD with a preserved vehicle

When preparing a pediatric omeprazole 2 mg/mL oral suspension using Ora-Plus® + Ora-Sweet® 1:1 as a preserved vehicle, what is the correct maximum BUD under USP <795> 2023?

- A)** 14 days (refrigerated)
- B)** 180 days (room temperature)
- C)** 35 days (refrigerated)
- D)** 90 days (room temperature)

ANSWER — QUICK QUESTION #1

Pediatric — BUD with a preserved vehicle

When preparing a pediatric omeprazole 2 mg/mL oral suspension using Ora-Plus® + Ora-Sweet® 1:1 as a preserved vehicle, what is the correct maximum BUD under USP <795> 2023?

- A) 14 days (refrigerated)
- B) 180 days (room temperature)
- C) 35 days (refrigerated) ✓ CORRECT**
- D) 90 days (room temperature)

Rationale: Ora-Plus® contains potassium sorbate (preservative). A preserved aqueous oral formulation → BUD 35 days per USP <795> 2023.

Clinical Case #1 — Pediatric

Ethosuximide oral suspension 50 mg/mL

Patient description

4-year-old child, 17 kg, with absence epilepsy. The neurologist prescribes Ethosuximide oral suspension 50 mg/mL — 200 mL, vanilla flavor. Commercial Zarontin® is not available locally → 503A compounding indicated.

Calculations + plan:

Dose: $20 \text{ mg/kg/day} \div \text{q12h} \rightarrow (17 \times 20) \div 2 = 170 \text{ mg/dose}$.

Volume: $170 \div 50 = 3.4 \text{ mL q12h}$ with an oral syringe.

Batch 200 mL: 10,000 mg ethosuximide + Ora-Blend® q.s. + 2 mL vanilla.

BUD: 35 days refrigerated.

Counseling: shake, refrigerate, do not freeze, warning signs (ataxia, drowsiness).

Clinical Case #2 — Pediatric

Clinical application with patient description

Patient description

8-month-old infant, 8 kg, with severe gastroesophageal reflux. The pediatrician prescribes: Omeprazole oral suspension 2 mg/mL — $0.7 \text{ mg/kg/dose} = 5.6 \text{ mg PO QD} \times 60 \text{ days}$. No commercial formulation is available at this exact dose → 503A compounding indicated.

Compounding plan

Calculation: Omeprazole 2 mg/mL in 50 mL = 100 mg API + Ora-Plus®/Ora-Sweet® 1:1.

Volume per dose: $5.6 \text{ mg} \div 2 \text{ mg/mL} = 2.8 \text{ mL}$ — use a calibrated oral syringe.

BUD: 14 days refrigerated (aqueous oral without preservative, USP <795> 2023).

Caregiver counseling: shake, oral syringe, refrigerate, do NOT mix into a full bottle.

Source: USP <795> 2023 · AAP 2023 · Allen's · PCCA Pediatric

QUICK QUESTION #2 · Learning Check

Pediatric Compounding — Caregiver Instructions

Patient: 3-year-old child. The mother receives a compounded oral suspension — she asks how to measure and give the dose.

Which instruction is BEST to ensure safe dosing?

- A) Use a kitchen teaspoon (~5 mL approximate).
- B) Use the calibrated oral syringe provided by the pharmacy.
- C) Mix the full dose into a bottle with juice.
- D) Estimate the dose by eye using a syringe from any pharmacy.

ANSWER — QUICK QUESTION #2

Pediatric Compounding — Caregiver Instructions

Which instruction is BEST to ensure safe dosing?

- A) Use a kitchen teaspoon (~5 mL approximate).
- B) Use the calibrated oral syringe provided by the pharmacy. ✓ CORRECT**
- C) Mix the full dose into a bottle with juice.
- D) Estimate the dose by eye using a syringe from any pharmacy.

Rationale: teaspoons and baby bottles dilute the dose and produce errors. The calibrated oral syringe is the pediatric standard (AAP 2023).

Dispensing Case — Enalapril Oral Solution

Pediatric Rx with dispensing calculations

Rx

Enalapril 1 mg/mL O.S.
Sig: 11 mg P.O. q.d. (2 mg PO once daily)
30 days of treatment

Question	Calculation / Answer
Quantity to dispense (30 days)	$2 \text{ mg/day} \div 1 \text{ mg/mL} \times 30 \text{ days} = 60 \text{ mL}$
Volume per daily dose	$2 \text{ mg} \div 1 \text{ mg/mL} = 2 \text{ mL}$ (NOT among 0.5/1/4/8 → "none of the above")
Storage label	Keep refrigerated (stable aqueous oral)
Additional label	"Shake well before use" + "Take 1 hour before meals"
Accessories to dispense	2 oz amber bottle + press-in adapter + 3 mL oral syringe

Source: USP <795> 2023 · Enalapril prescribing info · AAP Pediatric · ASHP

Veterinary Compounding (1/2)

AMDUCA + FDA-CVM regulatory framework (Obj 3)

AMDUCA (1994): permits extralabel use of FDA-approved drugs under a valid VCPR + a veterinary Rx + the absence of an adequate FDA product.

FDA-CVM: regulates veterinary compounding; prohibits copies of FDA-approved products (same as human 503A).

VCPR: documented and current Veterinarian-Client-Patient Relationship.

PR Act 247 + Regulation 156: the pharmacist must verify the veterinary license in PR and document the VCPR

Key veterinary calculations

- $\text{Dose} = \text{dose (mg/kg)} \times \text{animal weight (kg)}$.
- $\text{Volume/dose} = \text{Dose} \div \text{concentration}$.
- Canine example: Phenobarbital 2 mg/kg q12h, 30 kg Labrador → 60 mg/dose = 6 mL (10 mg/mL susp.). 30-day batch = 360 mL.

Source: AMDUCA 1994 · FDA-CVM 2023 · AVMA · PR Act 247-2004 · Plumb's

Veterinary Compounding (2/2)

Palatable dosage forms

Chewables (chews)

- Base: bovine gelatin + flavoring (liver, beef).
- Glycerin + water + flavoring.
- High owner adherence.
- APIs: phenobarbital, enalapril, metronidazole, gabapentin.

Flavored suspensions

- Ora-Plus + liver/beef flavoring.
- BUD 35 days refrigerated.
- Shake before use.
- Variable acceptance in cats.
- APIs: prednisolone, antibiotics.

Transdermal PLO/Lipoderm

- Solution for cats that refuse oral dosing.
- Application: inner ear pinna.
- APIs: methimazole, amitriptyline, ondansetron.
- **NEVER acetaminophen in cats.**

Source: AMDUCA · FDA-CVM · PCCA Veterinary · Allen's · Plumb's 10th Ed.

Species and Dose Adjustment

Comparison table of the main species

Species	Typical weight	Main challenge	Preferred form	Example API
Canine	1 – 70 kg	Wide dose range	Chews / suspension	Phenobarbital, enalapril
Feline	2 – 6 kg	Oral refusal	Transdermal PLO on the pinna	Methimazole TD
Equine	400 – 600 kg	Massive doses	Oral paste in a syringe	Omeprazole paste
Exotic (birds, reptiles)	0.05 – 5 kg	Minimal volumes	Concentrated solution	Varies by species

Feline-specific: NEVER acetaminophen · NEVER propylene glycol · extreme caution with NSAIDs. Deficient glucuronidation metabolism.

Source: AMDUCA · FDA-CVM · AVMA · Allen's · Carpenter's Exotic 5th Ed.

QUICK QUESTION #3 · Learning Check

Veterinary — AMDUCA Requirement

Under AMDUCA (1994) and FDA-CVM, which condition is NECESSARY for a pharmacist to prepare a compounded formulation for an animal?

- A)** A verbal request from the veterinarian without a written Rx.
- B)** A valid VCPR + a veterinary Rx + the absence of an adequate FDA-approved product.
- C)** Written authorization from the owner without a veterinary Rx.
- D)** A food-grade API certified for animal use.

ANSWER — QUICK QUESTION #3

Veterinary — AMDUCA Requirement

Under AMDUCA (1994) and FDA-CVM, which condition is NECESSARY for a pharmacist to prepare a compounded formulation for an animal?

- A) A verbal request from the veterinarian without a written Rx.
- B) A valid VCPR + a veterinary Rx + the absence of an adequate FDA-approved product. ✓
CORRECT**
- C) Written authorization from the owner without a veterinary Rx.
- D) A food-grade API certified for animal use.

Rationale: AMDUCA requires a documented VCPR + a valid veterinary Rx + clinical justification (no adequate FDA-approved product exists).

Clinical Case #3 — Veterinary

Phenobarbital liver-flavored suspension

Patient description

8 kg male canine (Beagle, 5 years) with idiopathic epilepsy. The veterinarian prescribes Phenobarbital liver-flavored oral suspension 10 mg/mL — 300 mL (30-day supply). VCPR documented. Valid veterinary Rx.

Calculations + dispensing

Dose: 2 mg/kg q12h $\rightarrow 2 \times 8 = 16$ mg/dose = 1.6 mL q12h.

Batch 300 mL: 3,000 mg phenobarbital + Ora-Plus 150 mL + Ora-Sweet 150 mL + liver FlavoRx 3 mL.

BUD: 35 days refrigerated.

Schedule IV: document the DEA log. Label: "For veterinary use only".

Owner counseling: oral syringe, ataxia/sedation = call the veterinarian.

Source: AMDUCA · FDA-CVM 2023 · Allen's · PCCA Veterinary · Plumb's 10th Ed.

BHRT — Fundamentals (1/2)

Definition + bioidentical hormones (Obj 4)

BHRT: hormones identical in molecular structure to endogenous human hormones (ovary, testis, adrenal gland).

vs. conventional HRT: MPA (Provera®) ≠ bioidentical progesterone · conjugated equine estrogens (Premarin®) ≠ bioidentical E1/E2/E3.

Main BHRT hormones

Estriol (E3): lower potency. Vaginal and urogenital atrophy. Vaginal route preferred.

Estradiol (E2): more potent. Vasomotor symptoms, osteoporosis. Transdermal, troche.

Estrone (E1): predominant after menopause. Oncologic risk debated.

Progesterone (P4): USP micronized. Uterine protection; oral, troche, vaginal, transdermal.

Testosterone: in women 0.5–2 mg/day TD. Libido, energy, muscle mass.

DHEA: adrenal precursor. Vaginal (Intrarosa®) or transdermal.

Source: NAMS 2022 · PCCA BHRT · ACOG 2022 · Allen's · Remington

BHRT — Fundamentals (2/2)

Bi-Est, Tri-Est and customized combinations

Combination	Typical ratio	Indication / Notes
Bi-Est (Biest) 80:20	E3 80% + E2 20%	Most common ratio — moderate vasomotor symptoms
Bi-Est 50:50	E3 50% + E2 50%	Greater estrogenic potency
Tri-Est	E3 80% + E2 10% + E1 10%	Less common — E1 implies a higher perceived risk
Biest + Progesterone	Customized	Troche or cream; uterine protection
Estradiol + Testosterone	Customized	TD gel; libido + vasomotor symptoms
Progesterone + DHEA	Customized	Transdermal cream; adrenal support
NAMS / ACOG position (2022): recognizes the use of compounded micronized progesterone. Recommends using FDA-approved products when available. Caution with unsupported safety claims.		

Source: NAMS 2022 · ACOG 2022 · PCCA BHRT · FDA BHRT Position

BHRT Dosage Forms

Comparison of 4 routes of administration

Form	Route	Advantages	Limitations
Troche	Sublingual / buccal	Rapid absorption, hepatic bypass	Variable absorption, taste
Oral capsule	Oral	Precise, BUD 180 days	Hepatic first pass
Transdermal cream	Skin (Topi-CLICK)	Flexible, non-hepatic, reproducible dose	Accidental transfer
Vaginal suppository	Vaginal	Direct local action, high local bioavailability	Compliance, messiness

Topi-CLICK®: each click = 0.25 mL. Sizes: 35mL is the most widely used, although they come in different sizes from 9mL up to 140mL. Reduces accidental transfer vs. jars. Ideal for TD BHRT.

Source: PCCA BHRT Guide · NAMS 2022 · Allen's · Topi-CLICK manufacturer · USP <795> 2023

BHRT Calculations

Formulas with examples for Biest, progesterone, testosterone

Bi-Est (Biest) 2.5 mg / 0.5 mL cream — 60 g batch (120 doses)

E3 (80%): $2.0 \text{ mg} \times 120 = 240 \text{ mg total}$ · E2 (20%): $0.5 \text{ mg} \times 120 = 60 \text{ mg total}$

Adjust each API individually for its CoA. Verify: total APIs / total volume = prescribed concentration.

Progesterone capsule 100 mg — 30 capsules

Total: $100 \times 30 = 3,000 \text{ mg}$. CoA adjustment 98.5% $\rightarrow 3,000 \div 0.985 = 3,045 \text{ mg}$ to weigh.

Excipient (Miglyol® / sunflower oil) q.s. to the #1 capsule fill weight (500 mg). BUD 180 days.

Topi-CLICK calibration verification

Each click = $0.25 \text{ mL} \approx 0.25 \text{ g}$ (density ≈ 1). Verification: weigh 5 consecutive clicks $\rightarrow \sim 1.25 \text{ g}$ expected.

If the deviation is >5%, recalibrate before dispensing.

BHRT Equipment and Devices

Critical: precision + absence of cross-contamination

Equipment / Device	Main function
Ointment mill	Homogenize API particles in semisolid creams; uniformity is critical for hormones
Analytical balance ± 0.1 mg	Hormones have a narrow therapeutic index — margin of error unacceptable
Topi-CLICK® (30/60/90/120 mL)	Calibrated TD dispensing device; each click = 0.25 mL
Troche molds	PVC / polypropylene, 1–2.5 mL/unit; lubricate before pouring PEG
Gelatin capsules #0–3	Oral progesterone: fill weight 350–450 mg in a #1 capsule
Mortar, spatulas, hotplate	Levigation, homogenization, PEG fusion

Mandatory cleaning between every BHRT formulation — risk of hormonal cross-contamination.

Source: USP <795> 2023 · PCCA BHRT Equipment · Topi-CLICK PI · Allen's

Testosterone — Dosing and Routes

BHRT in women and men · combinations

Women (BHRT)

- Dose: 0.5–2 mg/day transdermal.
- Indications: libido, energy, muscle mass, well-being.
- Common form: TD cream in Lipoderm + Topi-CLICK, 1 click QAM.
- Combined with Bi-Est or progesterone.
- Monitor: free testosterone, red blood cells, acne, hirsutism.

Men (Andropause)

- Dose: 50–100 mg/day transdermal (10× the female dose).
- FDA-approved: AndroGel®, Testim®, Axiron®.
- Compounding when commercial doses/routes do not apply.
- Compounded routes: troche, capsule, TD cream, scrotal.
- Monitor: total/free T, PSA, hematocrit, lipids.

Source: NAMS 2022 · ACOG 2022 · Endocrine Society Guidelines · PCCA BHRT

Penetration Enhancers

Increasing transdermal absorption in BHRT and dermatology

Function: they modify the stratum corneum or the partitioning of the API → greater TD flux.

Use judiciously: greater absorption = greater systemic exposure + possible irritation.

Enhancer	Typical concentration	Mechanism / Use
Propylene glycol (PG)	5 – 80%	Co-solvent; increases solubility and partitioning
DMSO (dimethyl sulfoxide)	10 – 50%	Lipid disruptor; restricted use due to irritation
Oleic acid	1 – 10%	Fluidizes the stratum corneum (lipophilic)
Ethanol	5 – 40%	Increases solubility; rapid evaporation
Urea	5 – 20%	Moisturizer + keratolytic — psoriasis
Menthol / terpenoids	1 – 5%	Reversible disruptor; pleasant sensation
Lipoderm® / PLO (PCCA)	Vehicle	Commercial systems with integrated enhancers

Source: USP <795> 2023 · PCCA Transdermal Guide · IJPC · Williams & Barry 2012

503A vs. 503B — Review (BHRT and Veterinary)

Regulatory application to Day 3

Day 3 context	Applicable DQSA section	Notes
Pediatric under an individual Rx	503A	PR Board of Pharmacy + USP <795>; no copies
Veterinary under VCPR + Rx	503A	Additionally AMDUCA + FDA-CVM; Schedule IV log if applicable
BHRT under an individual Rx	503A	Bi-Est, troches, customized TD — no FDA copies
Sterile product for a hospital	503B	Only an FDA-registered cGMP outsourcing facility
Anticipatory batches without an Rx	503B	Requires FDA registration + cGMP 21 CFR 210/211

Key rule for Day 3: Every pediatric, veterinary or BHRT formulation prepared in a PR community pharmacy operates under 503A — Board of Pharmacy + USP <795> + Act 247-2004 + AMDUCA (if veterinary).

Source: DQSA 2013 · FDA · AMDUCA · PR Act 247-2004 · NAMS 2022

Drug Shortage Exceptions

When compounding copies of FDA-approved products is allowed

General DQSA rule: 503A compounding may NOT prepare copies of FDA-approved products.

Exception: FDA Drug Shortage List — compounding is allowed while the product is in shortage.

Official source: accessdata.fda.gov/scripts/drugshortages/

Category	Documentation requirement
Prior verification	Confirm the product is on the FDA Drug Shortage List BEFORE preparing it
Documentation	Record the verification date and a screenshot/printout of the FDA page in the MFR
Discontinuation	Stop compounding when the shortage ends (periodic re-check)
Distribution	Maintain the $\leq 5\%$ interstate limit (503A) even during a shortage
Labeling	State "Compounded due to a drug shortage (FDA)" on the label

Source: DQSA 2013 Sec. 503A · FDA Drug Shortage Database · USP <795> 2023 · ASHP

Allergenic Excipient

Clinical justification for 503A reformulation

Justified case: the patient reacts to an excipient in the FDA-approved product but needs the API.

503A solution: prepare the API in a vehicle free of the allergenic excipient.

Common excipient	Reported reactions	Compounded alternative
Lactose	GI intolerance, milk allergy	Starch, MCC, mannitol
Gluten / wheat	Celiac disease, sensitivity	Corn/potato starch, gluten-free vehicles
Colorants (FD&C dyes)	Skin reactions, asthma	Clear/white capsules without colorant
Parabens / sulfites	Dermatitis, bronchospasm	Vehicles without paraben preservatives
Propylene glycol (PG)	Contact dermatitis	Lipoderm without PG, alternative vehicles
Sorbitol / saccharin	Osmotic diarrhea, phenylketonuria	Sucralose, stevia (pediatric)
Peanut / soy / lecithin	Anaphylaxis	Vehicles without soy lecithin

Document in the MFR: confirmed allergy, excluded excipient, alternative vehicle, communication with the prescriber.

Source: USP <795> 2023 · FDA Inactive Ingredients DB · AAP · PCCA · Allen's

Clinical Case #4 — BHRT

Bi-Est 2.5 mg/0.5 mL cream in Lipoderm — Topi-CLICK 60 g

Patient description

52-year-old woman with severe vasomotor symptoms (8–10 hot flashes/day, night sweats, insomnia, dyspareunia) 2 years post-menopause. No uterus (hysterectomy 2020). No history of breast cancer. The gynecologist prescribes Bi-Est 2.5 mg/0.5 mL TD cream in Lipoderm — 60 g in a Topi-CLICK 60 mL, 1 click every morning.

Batch calculations (60 g, 120 doses)

E3 (80%): $2.0 \text{ mg} \times 120 = 240 \text{ mg}$. CoA adjustment 99% → 242.4 mg.

E2 (20%): $0.5 \text{ mg} \times 120 = 60 \text{ mg}$. CoA adjustment 98% → 61.2 mg.

Lipoderm base q.s. to 60 g. Mix with an ointment mill (3 passes). Load into a Topi-CLICK 60 mL.

Verify: 5 consecutive clicks = ~1.25 g expected.

Counseling: rotate sites; cover until dry; avoid contact with partner/children; BUD 180 days.

Source: NAMS 2022 · PCCA BHRT · ACOG 2022 · USP <795> 2023 · Topi-CLICK PI

QUICK QUESTION #4 · Learning Check

BHRT — Route selection for urogenital atrophy

55-year-old patient with vaginal symptoms of urogenital atrophy. Which route maximizes LOCAL action while minimizing systemic absorption?

- A)** Sublingual estriol troche 2 mg
- B)** Transdermal Biest cream on the wrist
- C)** Oral estriol capsule 2 mg
- D)** Vaginal estriol cream or suppository

ANSWER — QUICK QUESTION #4

BHRT — Route selection for urogenital atrophy

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- B) Transdermal Biest cream on the wrist
- C) Oral estriol capsule 2 mg
- D) Vaginal estriol cream or suppository ✓ CORRECT**

Rationale: the vaginal route delivers estriol directly to the atrophic tissue. Minimal systemic absorption vs. oral/TD (NAMS 2022).

Application Activity — Laboratory

4 deliverable items (Day 3)

1

BHRT Cream — Bi-Est/Testosterone (LAB)

Bi-est 80/20 2.5 mg/g / Testosterone 3 mg/g cream — apply 0.5 g behind the knees QD.
Compounding Record + label.

2

Enalapril Suspension (LAB)

Enalapril 1 mg/mL — Dispense 60 mL, Sig 1 mL BID. Compounding Record + label + oral syringe.

3

Veterinary Gabapentin Suspension (LAB)

Gabapentin 100 mg/mL — Sig 1.5 mL PO q8h (Gabapentin 300 mg capsules + Ora-Plus/Ora-Sweet 1:1 q.s.). Compounding Record + veterinary label.

4

Synthroid 25 mcg/mL — Extemporaneous Preparation (LAB)

Conversion of Synthroid tablets (75 mcg Mon/Wed/Fri + 50 mcg Tue/Thu/Sat/Sun) into a 25 mcg/mL extemporaneous solution. Compounding Record + label.

Source: USP <795> 2023 · PCCA · Allen's — Practice formulations

Assessment

Post-test balanced across the 4 objectives

Post-Test structure: The Day 3 Post-Test covers the 4 objectives through 4 clinical cases (BHRT/Bi-est, Microclick, Enalapril, Gabapentin) in a balanced way:

- **Pre-test at the start** — Day 3 baseline.
- **Pop questions during the session** — 4 interspersed (pediatric ×2, veterinary, BHRT).
- **Case discussion** — 4 cases with a patient/animal description (pediatric ×2, veterinary, BHRT).
- **Lab report + calculations** — 4 practice formulations + Compounding Records (BHRT, Enalapril, veterinary Gabapentin, Synthroid).
- **Post-test** — 15 official questions, ≥ 70% for ACPE credits.
- **Distribution:** Q1–Q8 BHRT (Bi-est, Testosterone, Topi-CLICK, Microclick, enhancers) · Q9–Q13 Enalapril · Q14–Q15 veterinary Gabapentin.

Summary — Day 3

Six key points of the module

1

Pediatrics: vehicles (Ora-Plus/Sweet/Blend), oral syringe, BUD 35 days when preserved.

2

Pediatric cases: Ethosuximide (epilepsy) and Omeprazole (GERD) — full dose, volume, and BUD calculations.

3

Veterinary: AMDUCA + VCPR + Rx; species; mg/kg; chews / feline TD.

4

BHRT: Bi-Est E3/E2 ratios; troche, capsule, TD cream (Topi-CLICK), vaginal.

5

Calculations: CoA adjustment + concentration verification + Topi-CLICK calibration.

6

CPE: post-test $\geq 70\%$, 45 days, cfpr.org + CPE Monitor.

BHRT — 35 g Batch Calculation (test case)

Bi-est 80/20 + Testosterone — dispensing in a Topi-CLICK

Rx

Bi-est 80/20, 2.5 mg/G · Testosterone 3.0 mg/G
Disp. 35 G · Sig: apply 0.5 G behind the knees QD

Component	Calculation	Quantity for 35 G
Total estrogens (Bi-est)	$2.5 \text{ mg/G} \times 35 \text{ G}$	87.5 mg total
Estriol E3 (80%)	87.5×0.80	70 mg
Estradiol E2 (20%)	87.5×0.20	17.5 mg ('none of the above')
Testosterone	$3.0 \text{ mg/G} \times 35 \text{ G}$	105 mg ('none of the above')
Daily Topi-CLICK clicks	$0.5 \text{ G} \div 0.25 \text{ G/click}$	2 clicks QD
Reminder: Testosterone is Schedule III · the label requires the indication "compounded formulation" + Rx info + DEA log.		

Source: PCCA BHRT · USP <795> 2023 · DEA Schedule III · Allen's

Topi-CLICK vs. Microclick

Calibrated dispensing systems

Device	Capacities (G)	Volume per click	Typical use
Topi-CLICK®	30, 60, 90, 120	0.25 mL (≈ 0.25 G)	Standard BHRT, doses 0.25–2 mL
Microclick®	9.5	0.05 mL (≈ 0.05 G)	Very low doses: testosterone in women, pediatric DHEA

Calculation example (Day 3 test case)

- Rx:** Testosterone 1 mg/0.1 G in a Microclick. Sig: 1 mg QD.
- 1) Microclick capacity:** 9.5 G.
 - 2) One click =** 0.05 cc (≈ 0.05 G).
 - 3) Concentration:** $1\text{ mg} \div 0.1\text{ G} = 10\text{ mg/G}$. For a 12 G batch: $10 \times 12 = 120\text{ mg testosterone}$.
 - 4) Penetration enhancer scaling:** $5\text{ mL}/100\text{ mL} \times 12\text{ G} = 0.6\text{ mL}$.

Source: Topi-CLICK PI · Microclick PI · PCCA BHRT · Allen's

Penetration Enhancer — Selection

Propylene glycol as first choice in TD BHRT

For transdermal BHRT formulations (testosterone, Bi-est), the most appropriate and validated penetration enhancer is: **Propylene glycol (PG)**.

Enhancer	Primary indication	Notes
Propylene glycol (PG) ★	TD BHRT, steroids, antifungals	Most used and best tolerated; co-solvent
Ethoxy diglycol	Fragrances, some cosmetic APIs	Less common in BHRT — limited use
Versabase®	Vehicle / base — NOT a penetration enhancer per se	It is a vehicle, not an enhancer; test question
Glycerin	Humectant / co-solvent — NOT a primary enhancer	Different function; test question

Source: PCCA Penetration Enhancer Guide · IJPC · Williams & Barry 2012 · Allen's

Reconstitution ≠ Compounding

Key regulatory distinction

FDA / DQSA definition

- **Reconstituting an antibiotic (or any FDA-approved product) EXACTLY according to the manufacturer's instructions is NOT compounding.**
- It is "manipulation of a commercial product" under the FDA-approved labeling.
- It does not require 503A registration or USP <795> compliance for that step.

It IS compounding when...

- Proportions, concentration or dosage form are altered relative to the FDA product.
- Two or more APIs / products are combined.
- An excipient is omitted or substituted (e.g. allergy).
- A product is prepared from bulk API.

Source: DQSA Sec. 503A · FDA · USP <795> 2023 · PR Act 247-2004

USP Compliance ≠ Efficacy / Safety Claim

Regulatory clarification

FALSE: a pharmacist meeting USP standards in compounding is NOT thereby allowed to claim efficacy or safety of the compounded product.

Reason: compounded preparations do NOT undergo the clinical trials FDA requires for approved drugs. USP regulates the QUALITY of the process, not clinical EFFICACY/SAFETY.

What IS allowed

- Stating that the product meets USP <795>/<800> in process quality, documentation, BUD and equipment.
- Documenting that a validated Master Formulation Record (MFR) was followed.
- Telling the patient that it is NOT an FDA-approved product and describing its compounded nature.
- **ALWAYS labeling: "Compounded formula — not evaluated by FDA for safety or efficacy".**

Source: FDA Compounding Guidance · USP <795> 2023 · ACPE · DQSA

503A / 503B — Final Clarification

Common myth: they are NOT distinguished by sterile vs. non-sterile

MYTH: "503A compounds sterile products · 503B compounds non-sterile products."
X FALSE. The distinction is NOT sterility but the regulatory model.

Feature	503A	503B
Sterile compounding	Yes (under USP <797>)	Yes (under USP <797> + cGMP)
Non-sterile compounding	Yes (under USP <795>)	Yes (under USP <795> + cGMP)
Individual prescription	Required	Not required
Primary regulator	State Board of Pharmacy	FDA cGMP 21 CFR 210/211

Source: DQSA 2013 · FDA Compounding Guidance · USP <795>/<797>

POST-TEST

Day 3 · 15 official questions

Q1–Q6 BHRT (Bi-est, Testosterone, Topi-CLICK, Microclick) · Q7–Q8 Penetration Enhancers
Q9–Q13 Enalapril O.S. (quantity, volume, labels, accessories) · Q14–Q15 Gabapentin
(veterinary)

Passing score $\geq 70\%$ · One question per slide

POST-TEST · Question 1 / 15

BHRT Rx — Estriol

Patient: Rx: Bi-est 80/20, 2.5 mg/G · Testosterone 3.0 mg/G · Disp. 35 G · Sig: apply 0.5 G behind the knees QD · Refill 5. · Use this prescription to answer questions 1 through 3.

The amount of estriol required for this formulation is:

- A) 2.8 G
- B) 700 mg
- C) 70 mg
- D) 17.5 mg
- E) none of the above is correct

POST-TEST · Question 2 / 15

BHRT Rx — Testosterone

Patient: Rx: Bi-est 80/20, 2.5 mg/G · Testosterone 3.0 mg/G · Disp. 35 G · Sig: apply 0.5 G behind the knees QD · Refill 5.

The amount of testosterone required to prepare this formulation is:

A) 35 mg

B) 70 mg

C) 17.5 mg

D) 90 mg

E) none of the above is correct

POST-TEST · Question 3 / 15

Topi-CLICK Clicks

Patient: Rx: Bi-est 80/20, 2.5 mg/G · Testosterone 3.0 mg/G · Disp. 35 G · Sig: apply 0.5 G behind the knees QD · Refill 5.

If this product is dispensed in a Topi-CLICK, how many clicks per day should the patient apply?

- A) 0.5 click
- B) 1 click
- C) 2 click
- D) 5 click
- E) none of the above is correct

POST-TEST · Question 4 / 15

Microclick Capacity

Patient: Rx: Testosterone 1 mg/0.1 G · Disp. in a microclick · Sig: 1 mg QD. · Use this prescription to answer questions 4 through 8.

What is the capacity of the microclick in G?

A) 9 G

B) 9.5 G

C) 30 G

D) 35 G

E) none of the above is correct

POST-TEST · Question 5 / 15

Microclick Volume

Patient: Rx: Testosterone 1 mg/0.1 G · Disp. in a microclick · Sig: 1 mg QD.

When you make one microclick, how much of the preparation is dispensed?

A) 0.025 cc

B) 0.5 cc

C) 0.05 cc

D) 0.1 cc

E) none of the above is correct

POST-TEST · Question 6 / 15

Testosterone — Calculation with an Excess

Patient: Rx: Testosterone 1 mg/0.1 G · Disp. in a microclick · Sig: 1 mg QD.

How much testosterone is needed to prepare this formulation if you decide to make an excess of 12 G?

A) 1.2 mg

B) 12 mg

C) 120 mg

D) 10 mg

E) none of the above is correct

POST-TEST · Question 7 / 15

Penetration Enhancer — Selection

Patient: Rx: Testosterone 1 mg/0.1 G · Disp. in a microclick · Sig: 1 mg QD.

For this transdermal testosterone formulation, which will be the most appropriate penetration enhancer?

- A) ethoxy diglycol
- B) propylene glycol
- C) versabase
- D) glycerin
- E) none of the above is correct

POST-TEST · Question 8 / 15

Penetration Enhancer — Calculation

Patient: Rx: Testosterone 1 mg/0.1 G · Disp. in a microclick · Sig: 1 mg QD.

The formula to make 100 mL requires 5 mL of the penetration enhancer. If you decide to make only 12 G of the formula, how much penetration enhancer will you use?

A) 0.5 mL

B) 0.6 mL

C) 5 mL

D) 6 mL

E) none of the above is correct

POST-TEST · Question 9 / 15

Enalapril Rx — Quantity

Patient: Rx: Enalapril 1 mg/mL O.S. · Sig: II mg PO QD (2 mg PO once daily) · 30 days. · Use this prescription to answer questions 9 through 13.

What quantity will you dispense?

A) 30 mL

B) 60 mL

C) 90 mL

D) 180 mL

E) none of the above is correct

POST-TEST · Question 10 / 15

Enalapril Rx — Daily Volume

Patient: Rx: Enalapril 1 mg/mL O.S. · Sig: II mg PO QD (2 mg PO once daily) · 30 days.

How many mL will the patient have to take per day?

A) 0.5 mL

B) 1 mL

C) 4 mL

D) 8 mL

E) none of the above is correct

POST-TEST · Question 11 / 15

Enalapril Rx — Storage Label

Patient: Rx: Enalapril 1 mg/mL O.S. · Sig: II mg PO QD (2 mg PO once daily) · 30 days.

Regarding storage, which label will you place on the Enalapril suspension?

- A) keep frozen
- B) keep at room temperature
- C) keep in the safety cabinet
- D) keep in the refrigerator
- E) none of the above is correct

POST-TEST · Question 12 / 15

Enalapril Rx — Additional Label

Patient: Rx: Enalapril 1 mg/mL O.S. · Sig: II mg PO QD (2 mg PO once daily) · 30 days.

Which other label will you place?

- A) shake well before use
- B) take 1 hour before meals
- C) take 1 hour after meals
- D) A and B are correct
- E) none of the above is correct

POST-TEST · Question 13 / 15

Enalapril Rx — Accessories

Patient: Rx: Enalapril 1 mg/mL O.S. · Sig: II mg PO QD (2 mg PO once daily) · 30 days.

What else will you provide with this prescription?

- A) press-in adapter
- B) two-ounce amber bottle
- C) 3 mL oral syringe
- D) all of the above
- E) none of the above is correct

POST-TEST · Question 14 / 15

Gabapentin Rx — Quantity to Dispense

Patient: Rx: Gabapentin 100 mg/mL · Sig: 1.5 mL PO q8h. · Use this prescription to answer questions 14 through 15.

The quantity to prepare in order to dispense this prescription is:

A) 30 mL

B) 60 mL

C) 135 mL

D) 135 mg

E) none of the above is correct

POST-TEST · Question 15 / 15

Gabapentin Rx — Quantity per Dose

Patient: Rx: Gabapentin 100 mg/mL · Sig: 1.5 mL PO q8h.

The amount of gabapentin per dose this patient will receive is:

A) 150 mg

B) 100 mg

C) 150 mL

D) 100 mL

E) none of the above is correct

ANSWER KEY — POST-TEST (1 of 2)

For facilitator use / participant self-assessment · Questions 1–8

Q#	Answer	Rationale
Q1	C	Total estrogen = $2.5 \text{ mg/G} \times 35 \text{ G} = 87.5 \text{ mg}$; at the 80:20 Bi-est ratio, estriol (E3) = $87.5 \text{ mg} \times 0.80 = 70 \text{ mg}$.
Q2	E	Testosterone = $3.0 \text{ mg/G} \times 35 \text{ G} = 105 \text{ mg}$, which is not among the listed options (A–D) — so the correct choice is E.
Q3	C	The Sig calls for 0.5 G/day; each Topi-CLICK click delivers 0.25 G, so $0.5 \div 0.25 = 2$ clicks per day.
Q4	B	The standard Microclick® device has a 9.5 G capacity — much smaller than the 30/60/90/120 mL Topi-CLICK sizes, for very low-dose formulations.
Q5	C	Each Microclick delivers 0.05 mL ($\approx 0.05 \text{ g}$) per click — one-fifth the volume of a standard Topi-CLICK click (0.25 mL).
Q6	C	Concentration = $1 \text{ mg} \div 0.1 \text{ G} = 10 \text{ mg/G}$; for a 12 G batch, $10 \text{ mg/G} \times 12 \text{ G} = 120 \text{ mg}$ of testosterone.
Q7	B	Propylene glycol is the most used and best-tolerated penetration enhancer for transdermal BHRT/steroid formulations; DMSO is too irritating, and Versabase/glycerin are vehicles, not enhancers.
Q8	B	The formula uses 5 mL of enhancer per 100 mL; scaled to a 12 G batch: $(5 \text{ mL} \div 100) \times 12 = 0.6 \text{ mL}$.

ANSWER KEY — POST-TEST (2 of 2)

For facilitator use / participant self-assessment · Questions 9–15

Q#	Answer	Rationale
Q9	B	$2 \text{ mg/day} \div 1 \text{ mg/mL} = 2 \text{ mL/day}$; $2 \text{ mL/day} \times 30 \text{ days} = 60 \text{ mL}$ to dispense.
Q10	E	$2 \text{ mg} \div 1 \text{ mg/mL} = 2 \text{ mL/day}$, which is not among the listed options (A–D) — so the correct choice is E.
Q11	D	This is an aqueous oral suspension with no preservative noted, so it is labeled to keep refrigerated to protect stability and limit microbial growth.
Q12	D	Both labels apply: shake well before use (uniform suspension) and take 1 hour before meals (per enalapril administration instructions).
Q13	D	A complete dispensing includes the press-in adapter, the 2 oz amber bottle (light protection), and a 3 mL oral syringe for accurate dosing.
Q14	C	At 1.5 mL q8h (3 doses/day) for the standard 30-day supply used throughout this course: $1.5 \text{ mL} \times 3 \times 30 \text{ days} = 135 \text{ mL}$.
Q15	A	At 100 mg/mL, a 1.5 mL dose contains $1.5 \times 100 \text{ mg} = 150 \text{ mg}$ of gabapentin.

Each question answered correctly represents demonstration of achievement of the corresponding objective.

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Thank you for your participation!

Day 3 completed — Basic Non-Sterile Compounding · Course finished

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